

Date: Feb 9, 2021

Experiment 5

Title

Preparation of aspirin (Acetylsalicylic acid)

Objective

The objective of this experiment was to prepare acetyl salicylic acid, commonly known as "Aspirin"

Theory

Aspirin also known as acetylsalicylic acid (ASA): is a medication used to reduce pain, fever, or inflammation. To prepare aspirin, salicylic acid is reacted with excess of acetic anhydride. Small amount of a strong mineral acid or a weak base can be used as a catalyst. To speed up the reaction it is necessary to warm the reaction mixture using gentle heat such as placing the mixture over a water bath.

Procedure

Preparation of aspirin

1gm of salicylic acid was measured and prepared by our instructor; this amount was put into a dry test tube. 2ml of acetic anhydride was added to this test tube, and then following that we added 5 drops of conc. Sulphuric acid. Water was added to a water bath, and heated by a Bunsen burner. When it started to warm, the test tube containing the salicylic acid, acetic anhydride and sulphuric acid was partially immersed into the hot water bath. Using a stirrer the mixture was stirred. 30ml of water was measured using a measuring cylinder, and added to a beaker. The mixture was added to the beaker containing the 30ml. The beaker was placed into an ice bath for the process of crystal formation. After a few minutes crystals started forming slowly. A suction filter was assembled, and a filter paper was put on top. The mixture that formed crystals was transferred into the suction filter. The product was left on the filter paper. It was dried using paper for the next reactions of solubility.

Solubility test of aspirin

After drying the product we filtered in the previous reaction, the resulting product was collected. 3 dry test tubes were prepared. 1ml of Water, ethanol, and sodium bicarbonate was added into the three different test tubes. A small amount of the product aspirin was put on to the tip of a spatula and it was added to the three different test tubes. The results of each was observed and recorded.

Solubility test of salicylic acid

3 dry test tubes were prepared. 1ml of Water, ethanol, and sodium bicarbonate was added into the three different test tubes. A small amount of salicylic acid was put on to the tip of a spatula and it was added to the three different test tubes. The results of each was observed and recorded.

Test for phenolic hydroxyl group

1 drop of 1% ferric chloride solution was added to the above test tubes that were used to test for solubility. The result was observed and recorded.

Apparatus used

- Water bath
- Bunsen burner
- Clamp
- Boss head
- Suction filter
- Filter paper
- Test tube
- Stand
- Beaker
- Test tube holder
- Spatula
- Stirrer
- Measuring cylinder

Chemical used

- Water
- Ethanol
- Sodium bicarbonate
- Conc. Sulphuric acid
- Acetic anhydride
- Aspirin

Result and discussion

	solvents		
sample	water	ethanol	NaHCO ₃
Aspirin	Not soluble	soluble	Soluble
Salicylic acid	Not soluble	soluble	soluble

The experiment result shows that the solubility of Aspirin(Acetyl salicylic acid) in ethanol and NaHCO_3 is high when compared to water. Aspirin has a limited solubility in water, the hydrophobic nature of aspirin makes it insoluble.

Salicylic acid is soluble in water, and it releases carbon dioxide.

	Solvent		
Sample	water	ethanol	NaHCO_3
Aspirin	Reaction(from colorless to purple)	Reaction(from colorless to light purple)	Reaction(from colorless to yellow)
Salicylic acid	Reaction(from colorless to purple)	Reaction(from colorless to purple)	Reaction(from colorless to pink/orange)

When ferric chloride solution is added to a test tube that contains Aspirin and water, it changes the color from colorless to purple, when it is added to Aspirin and NaHCO_3 solution it changes its color from colorless to yellow. **However when we add ferric chloride in Aspirin and ethanol, it changes its color from colorless to light purple, but this result should have been colorless. The result recorded on the table occurred due to the contact between the dropper that was previously used for salicylic acid.**

When ferric chloride solution is added to a test tube containing salicylic acid and ethanol also salicylic acid and water, both change their color from colorless to purple. However when we add it to a solution of salicylic acid and NaHCO_3 it changes its color from colorless to pink/orange.

Quiz questions

- Why is the carboxyl group of salicylic acid not acetylated?
 - During the synthesis of aspirin, the carboxylic group of salicylic acid is not acetylated; this is because of the lower nucleophilicity of the carboxylic group. That's why acetic anhydride prefers phenoic-OH group rather than carboxylic group. Carboxylic acids are not as nucleophilic as phenols. It is possible to form a mixed anhydride of a carboxylic acid, but likely the conditions used were either not strong enough or it was formed and was hydrolyzed upon workup (mixed anhydrides are also not too stable).
- What is the role of the acid catalyst? Also a base like pyridine can be used for the catalysis. Explain the role of the catalyst by giving the mechanism of the reaction.
 - Just a few drops of sulphuric acid added to the mixture means that there are some free positive hydrogen ions in the solution which can bind to the acetic anhydride. The acid serves to protonate one double bonded oxygen on the acetic anhydride this makes the reaction with the salicylic acid faster. The proton is returned to the acid in the end so it is just a catalyst.
 - Pyridine is more nucleophilic than the alcohol, and it attacks the acyl chloride rapidly, forming a highly electrophilic (because of the positive charge) intermediate. Because pyridine is acting as a nucleophile to speed up the reaction, yet is unchanged by the reaction, it is a "nucleophilic" catalyst.

3. The melting point of salicylic acid is 157°C-159°C while that of aspirin, which has a higher molecular formula, is around 135°C. Explain why?
- The Carboxylic acid group in aspirin's structure allows molecules of aspirin to form hydrogen bonds with each other. Because of the permanent dipole bonds within the aspirin molecule. Because of such a strong attraction, it indicates that aspirin's melting point is relatively high. **However, the melting point of salicylic acid is expected to be higher than the melting point of aspirin as there are more hydrogen bonds formed which means that more energy is needed to break the bonds. So when testing the purification of aspirin, pure salicylic acid's melting point should be higher than the aspirin sample.**
4. What is the rationale for the observed color reaction between salicylic acid and ferric chloride?
- FeCl_3 (aq). Iron (III) ion reacts with phenols to form a purple complex. Salicylic acid contains a phenol group, but acetylsalicylic acid does not. Therefore, if you add FeCl_3 to an aspirin sample and you see a purple color, it means that there is still some salicylic acid present and the sample is impure. FeCl_3 will not react with aspirin. It will, however, react with salicylic acid, which is used to synthesize aspirin. FeCl_3 (aq). Iron (III) ion reacts with phenols to form a purple complex. Salicylic acid contains a phenol group, but acetylsalicylic acid does not. A purple colour is an indication of salicylic acid. A sample of pure aspirin should not exhibit a colour change.



COLLEGE OF HEALTH SCIENCE

DEPARTMENT: PHARMACY

Group 4

ORGANIC CHEMISTRY LAB

REPORT: EXP 5

GROUP MEMBERS:

- 1. MERON ASFAW: UGR/0336/12**
- 2. MEZID MUHAMMED: UGR/9636/12**
- 3. MELAT SALLE:UGR/1574/12**

SUBMITTED TO: INSTUCTOR KIBROM GEBREHIWOT

DATE OF SUBMISSION: FEB 15, 2021